

Junyi Zhu

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ABOUT ME

Senior Researcher at Samsung Research UK with a PhD in Artificial Intelligence from KU Leuven. Experienced in generative AI—including large language models and diffusion models—as well as efficient training and inference algorithms and privacy-preserving machine learning. Author of multiple papers at leading AI conferences such as NeurIPS, ICML, ICLR, CVPR, and EMNLP. Passionate about advancing AI systems from research to real-world applications, bridging cutting-edge methods with scalable, impactful solutions.

SKILLS

MACHINE LEARNING & AI

Large Language Models (LLM); Diffusion Models; Text and Image Embedding Models; Reinforcement Learning (RL); Retrieval-Augmented Generation (RAG); Federated Learning; Differential Privacy (DP)

LANGUAGE

English; Chinese; German

WORK EXPERIENCE

SAMSUNG RESEARCH UK

SENIOR RESEARCHER

London, UK • Sep 2024 – Present

- Led research and prototyping of mobile language models for personalized on-device AI systems.
- Designed and implemented scalable federated training pipelines for privacy-preserving personalization.

KU LEUVEN, ESAT-PSI LAB

RESEARCH ASSOCIATE

Leuven, Belgium • Apr 2024 – Sep 2024

- Researched AI search systems and efficiency of training and inference in image generative models.

EXTERNAL RESEARCH COLLABORATIONS

SCIENTIFIC COLLABORATOR

Global • Oct 2023 – Present

- **Microsoft Research (Redmond)**: Collaborated on improving diffusion-based image generation and developing unified principles for generative modeling, representation learning, and classification.
- **MemTensor**: Post-training large language models with reinforcement learning and building AI search systems based on retrieval-augmented generation (RAG).
- **Infinigence-AI**: Worked on improving training and inference efficiency for large language models and diffusion models.

HELLA KGAA HUECK & CO.

RESEARCH INTERN

Lippstadt, Germany • Nov 2017 – Feb 2018

- Developed a data analysis framework for driver-assistance radar sensors.

ACADEMIC ENGAGEMENT

AREA CHAIR

REVIEWER

VISITING SCHOLAR

CVPR 2026
ICML, NeurIPS, ICLR, CVPR, ACL, EMNLP, ICCV, TMLR (2021–2025)
NICS-EFC Lab, Tsinghua University (Aug–Oct 2023)

TALKS (SELECTED)

ICLR TINY PAPER TRACK (2024)

RESCALING INTERMEDIATE FEATURES MAKES TRAINED CONSISTENCY MODELS PERFORM BETTER

LEUVEN.AI JUNIOR RESEARCHERS DAY (2023)

NEURIPS WORKSHOP (2022)

TACKLING PERSONALIZED FEDERATED LEARNING WITH LABEL CONCEPT DRIFT

SELECTED PUBLICATIONS

*FULL LIST ON GOOGLE SCHOLAR

- Wang, Z. *, **Zhu, J. ***, Tang, B., Li, Z., et al. *Jigsaw-R1: A Study of Rule-based Visual Reinforcement Learning with Jigsaw Puzzles*. arXiv, 2025. (*co-first*)
- Lin, Z., Liu, E., Ning, X., **Zhu, J.**, et al. *Latent Zoning Network: A Unified Principle for Generative Modeling, Representation Learning, and Classification*. NeurIPS, 2025.
- **Zhu, J.**, Ozkan, S., Maracani, A., Mutlu, S., et al. *Multi-Task Pre-Finetuning of Lightweight Transformer Encoders for Text Classification and NER*. EMNLP, 2025.
- Wang, Y., Wang, P., Xi, C., Tang, B., **Zhu, J.**, et al. *Adversarial Preference Learning for Robust LLM Alignment*. ACL Findings, 2025.
- Tang, B. *, **Zhu, J. ***, Wang, H., Xi, C., et al. *Xinyu AI Search: Enhanced Relevance and Comprehensive Results with Rich Answer Presentations*. arXiv, 2025. (*co-first*)
- Liu, E. *, **Zhu, J. ***, Lin, Z., Ning, X., et al. *Linear Combination of Saved Checkpoints Makes Consistency and Diffusion Models Better*. ICLR, 2025. (*co-first*)
- **Zhu, J.**, Liu, S., Yu, Y., Tang, B., et al. *FastMem: Fast Memorization of Prompt Improves Context Awareness of Large Language Models*. EMNLP Findings, 2024. (*co-first*)
- Liu, E. *, **Zhu, J. ***, Lin, Z., Ning, X., et al. *Efficient Expert Pruning for Sparse Mixture-of-Experts Language Models: Enhancing Performance and Reducing Inference Costs*. arXiv, 2024. (*co-first*)
- **Zhu, J.**, Lin, Z., Liu, E., Ning, X., Blaschko, M. B. *Rescaling Intermediate Features Makes Trained Consistency Models Perform Better*. ICLR Tiny Papers (Notable), 2024.
- **Zhu, J.**, Yao, R., Blaschko, M. B. *Surrogate Model Extension: A Fast and Accurate Weight Update Attack on Federated Learning*. ICML, 2023.
- **Zhu, J.**, Blaschko, M. B. *Improving Differentially Private SGD via Randomly Sparsified Gradients*. TMLR, 2023.
- **Zhu, J. ***, Ma, X. *, Blaschko, M. B. *Confidence-aware Personalized Federated Learning via Variational Expectation Maximization*. CVPR, 2023. (*co-first*)
- **Zhu, J.**, Blaschko, M. B. *R-GAP: Recursive Gradient Attack on Privacy*. ICLR, 2021.

EDUCATION

KU LEUVEN

PH.D., ARTIFICIAL INTELLIGENCE

Leuven, Belgium

- Dissertation: Privacy-Preserving and Distributed Machine Learning
- Funded by Flemish Government (AI Research Program) and FWO Scholarship

KARLSRUHE INSTITUTE OF TECHNOLOGY

M.Sc., INFORMATION TECHNOLOGY AND VEHICLE ENGINEERING

Karlsruhe, Germany

- Thesis: Transferring Grid Maps to Aerial Imagery for Localization in Autonomous Driving using Generative-Adversarial Networks (GANs)

DRESDEN UNIVERSITY OF TECHNOLOGY

B.Sc. (EXCHANGE)

Dresden, Germany

- Thesis: Design and Implementation of Automated Penetration Measuring System

BEIJING INSTITUTE OF TECHNOLOGY

B.Sc., MECHANICAL ENGINEERING AND AUTOMATION

Beijing, China